

AKSA-SDS Summer Internship

Name: Laiba Ishaq

Dept: AI

Introduction

This report presents a detailed study of unsupervised machine learning and model interpretation techniques, based on a practical implementation carried out in Python using scikit-learn, umap-learn, and SHAP. Two datasets were used to cover both a low-dimensional clustering scenario and a high-dimensional dimensionality-reduction scenario, as required by the task. The first dataset, the Wholesale Customers dataset, was used to explore clustering algorithms on a moderately sized, low-dimensional tabular dataset. The second dataset, the Olivetti Faces dataset, was used to explore dimensionality reduction and clustering on a genuinely high-dimensional image dataset, and subsequently to demonstrate model interpretation techniques using a supervised proxy model.

The report is organised into three main parts. The first part explains the theoretical concepts underlying unsupervised learning, illustrated with concrete examples drawn from the notebook. The second part applies at least five clustering or dimensionality reduction algorithms to each dataset, compares their performance using standard evaluation metrics, tunes their hyperparameters, and visualises the resulting clusters before and after dimensionality reduction. The third part interprets the best-performing model using feature importance, permutation importance, SHAP, and partial dependence plots. All figures, tables, and numerical results reported here are reproduced exactly as they were generated by the notebook; no values have been invented or approximated beyond the rounding already applied in the original code.

Theoretical Foundations of Unsupervised Learning

Introduction to Unsupervised Learning

Unsupervised learning refers to a class of machine learning methods that discover structure in data without relying on labelled outcomes. Rather than learning a mapping from inputs to known outputs, an unsupervised algorithm looks for patterns, groupings, or a more compact representation of the data on its own. In this project, the Wholesale Customers dataset was treated as unlabeled: the notebook did not use the existing Channel or Region columns as targets for the clustering task, but instead let clustering algorithms discover natural groupings of customers purely from their annual spending on Fresh, Milk, and Grocery products. Similarly, the Olivetti Faces dataset, although it technically has a Person_ID label, was projected and clustered without using that label during model fitting; the label was only used afterwards to evaluate how well the discovered clusters aligned with the true identities.

Clustering vs Dimensionality Reduction

Clustering and dimensionality reduction are the two main branches of unsupervised learning explored in this report, and although they are often used together, they solve different problems. Clustering partitions the data points themselves into groups such that points within a group are more similar to each other than to points in other groups; the output is a discrete label per observation. Dimensionality reduction, by contrast, transforms the feature space itself, producing

a smaller number of derived features that retain as much of the meaningful structure of the data as possible; the output is a new, lower-dimensional representation of every observation. In the notebook, K-Means, Hierarchical Clustering, DBSCAN, GMM, and Mean Shift are examples of clustering, while PCA, Kernel PCA, t-SNE, and UMAP are examples of dimensionality reduction. The two techniques were also combined directly: clustering algorithms were re-applied on top of PCA-reduced and Kernel-PCA-reduced data to see whether reducing dimensionality first improved the quality of the discovered clusters.

Distance Metrics

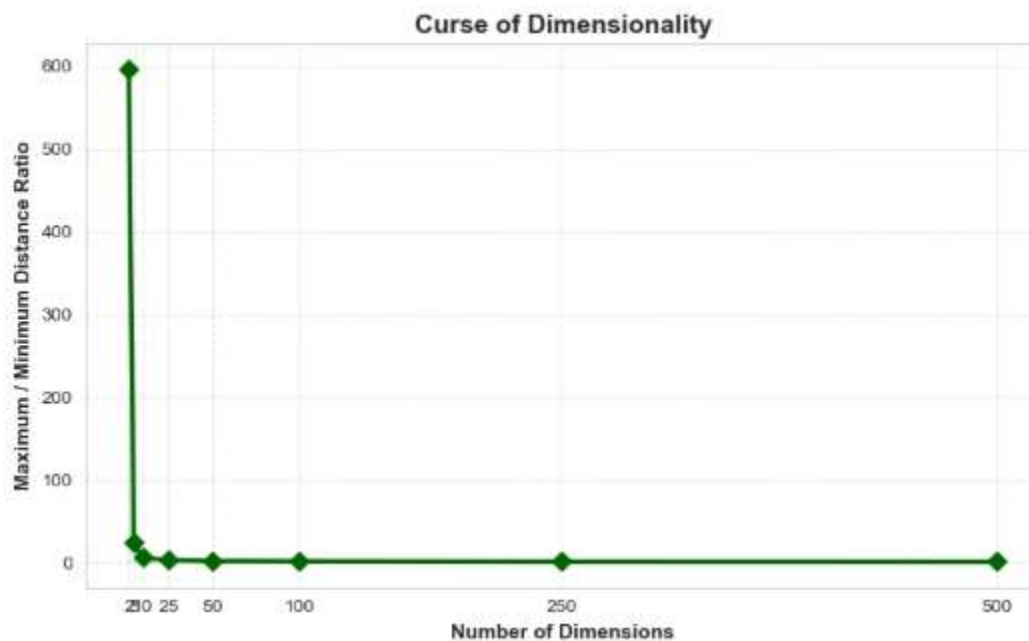
Because most clustering algorithms rely on some notion of how far apart two data points are, the choice of distance metric materially affects the resulting clusters. The notebook computed three standard distance metrics between the first two customers in the Wholesale Customers dataset (before any scaling was applied): the Euclidean distance, the Manhattan (city-block) distance, and the Cosine distance. The exact output produced was a Euclidean distance of 5962.07, a Manhattan distance of 7773, and a Cosine distance of 0.0563. The large numeric gap between the Euclidean/Manhattan values and the Cosine value illustrates an important point: Euclidean and Manhattan distances are sensitive to the absolute magnitude of the raw spending figures, which for these two customers differ substantially across Fresh, Milk, and Grocery. Cosine distance, on the other hand, measures the angle between the two spending vectors rather than their magnitude, so it reflects how similar the proportional spending pattern of the two customers is regardless of how much they spend overall; the low Cosine distance of 0.0563 indicates that, proportionally, these two customers spend in a fairly similar pattern across the three product categories even though their absolute totals differ.

Feature Scaling for Clustering

Distance-based algorithms such as K-Means, DBSCAN, and Mean Shift implicitly assume that all features contribute comparably to the notion of distance. When features are measured on very different scales, as is the case for Fresh, Milk, and Grocery spending (which range from single digits to over one hundred thousand monetary units), the feature with the largest numeric range dominates the distance calculation and effectively determines the clustering outcome on its own. To correct this, the notebook applied a StandardScaler, which rescales every feature to zero mean and unit variance. After scaling, the same three features for the first five customers were transformed into small, comparable values, for example the first customer's scaled Fresh, Milk, and Grocery values became approximately 0.0529, 0.5236, and -0.0411 respectively, rather than the original values of 12669, 9656, and 7561. This ensures that no single product category numerically overwhelms the clustering process purely because of its larger raw scale. The notebook later revisited this decision explicitly in a dedicated preprocessing experiment, comparing StandardScaler against MinMaxScaler, RobustScaler, and no scaling at all.

Curse of Dimensionality

The curse of dimensionality describes the phenomenon whereby, as the number of features describing each data point grows, distance-based notions of similarity become progressively less meaningful. In very high-dimensional spaces, the difference between the nearest and farthest neighbour of any given point tends to shrink relative to the average distance, so that all points start to look almost equally far apart. The notebook demonstrated this directly with a synthetic experiment: for a fixed sample of three hundred random points, the ratio of the maximum pairwise distance to the minimum pairwise distance was computed as the number of random dimensions was increased through the values 2, 5, 10, 25, 50, 100, 250, and 500.



As the plotted curve shows, this ratio is largest when the data has very few dimensions and steadily collapses toward one as dimensionality increases, confirming that in very high-dimensional spaces, distances between random points become nearly indistinguishable. This result directly motivated the treatment of the Olivetti Faces dataset later in the report, which starts with 4096 raw pixel features for only 400 images; without some form of dimensionality reduction or feature selection, distance-based clustering on such data is expected to perform poorly, which is indeed what was observed.

Principal Component Analysis (PCA) – Overview

Principal Component Analysis is a linear dimensionality-reduction technique that re-expresses the data along new orthogonal axes, called principal components, ordered so that the first component captures the greatest possible variance in the data, the second component the next greatest amount of variance orthogonal to the first, and so on. Because a small number of components can often capture most of the variance in the original features, PCA is commonly used both to compress

high-dimensional data for visualisation and to remove redundant or correlated information before clustering. PCA is examined in detail in the high-dimensional dataset analysis in Section 4, where it is also compared against Kernel PCA, t-SNE, and UMAP.

t-SNE vs UMAP

t-Distributed Stochastic Neighbour Embedding (t-SNE) and Uniform Manifold Approximation and Projection (UMAP) are both non-linear dimensionality-reduction techniques designed primarily for visualisation of high-dimensional data in two or three dimensions. t-SNE converts pairwise similarities in the original high-dimensional space and in the low-dimensional embedding into probability distributions, and then minimises the divergence between the two, which tends to produce visually well-separated clusters but can distort global distances between clusters and is sensitive to the perplexity hyperparameter, which loosely controls the effective number of neighbours considered around each point. UMAP is based on ideas from topological data analysis and manifold learning; it tends to run faster than t-SNE, and is generally considered better at preserving both local neighbourhood structure and a degree of global structure between clusters. Its main hyperparameters are the number of neighbours, which plays a similar role to t-SNE's perplexity, and the minimum distance, which controls how tightly points are allowed to be packed together in the embedding. Both methods were applied to the Olivetti Faces dataset in Section 4, including a hyperparameter comparison for perplexity (t-SNE) and the number of neighbours (UMAP).

Cluster Validation Techniques

Since unsupervised clustering has no ground-truth labels to check against directly, cluster quality is instead assessed using internal validation metrics that quantify how compact and well separated the discovered clusters are. This report uses four such metrics throughout: the Silhouette Score, the Davies-Bouldin Index, the Calinski-Harabasz Index, and Inertia. These are described in detail in Section 2.9 to 2.12 below and are applied consistently to every clustering algorithm and dimensionality-reduction method examined in Sections 3 and 4.

Cluster Evaluation Metrics

Silhouette Score

The Silhouette Score measures, for each point, how similar it is to its own cluster compared to the next-nearest cluster, producing a value between -1 and 1. Values close to 1 indicate that points are well matched to their own cluster and clearly separated from neighbouring clusters, values near 0 indicate overlapping clusters, and negative values suggest points may have been assigned to the wrong cluster. This was the primary metric used in the notebook to select the optimal number of clusters for K-Means and to rank clustering algorithms and dimensionality-reduction methods against each other.

Davies-Bouldin Index

The Davies-Bouldin Index summarises, for every pair of clusters, the ratio of within-cluster scatter to between-cluster separation, and averages the worst such ratio for each cluster. Lower values indicate better clustering, since they correspond to clusters that are compact internally and far apart from one another. This metric is particularly useful as a cross-check against the Silhouette Score, since the two are computed differently and do not always agree.

Calinski-Harabasz Index

The Calinski-Harabasz Index, also known as the Variance Ratio Criterion, compares the between-cluster dispersion to the within-cluster dispersion, scaled by the number of clusters and data points. Higher values indicate better-defined clusters. Because this metric is sensitive to the absolute scale of the data, it should generally be interpreted alongside the Silhouette Score and Davies-Bouldin Index rather than in isolation, a point that becomes especially relevant in the Kernel PCA results discussed in Section 4.

Inertia (Within-Cluster Sum of Squares)

Inertia is the sum of squared distances of every point to its assigned cluster centroid, and is the objective function that K-Means directly minimises. It always decreases as the number of clusters increases, so it is not by itself a fair criterion for choosing the number of clusters; instead, it is typically examined visually for an 'elbow', the point beyond which adding further clusters yields diminishing reductions in inertia. The notebook used inertia together with the Silhouette Score for this purpose.

Datasets Used

Wholesale Customers Dataset (Clustering Dataset)

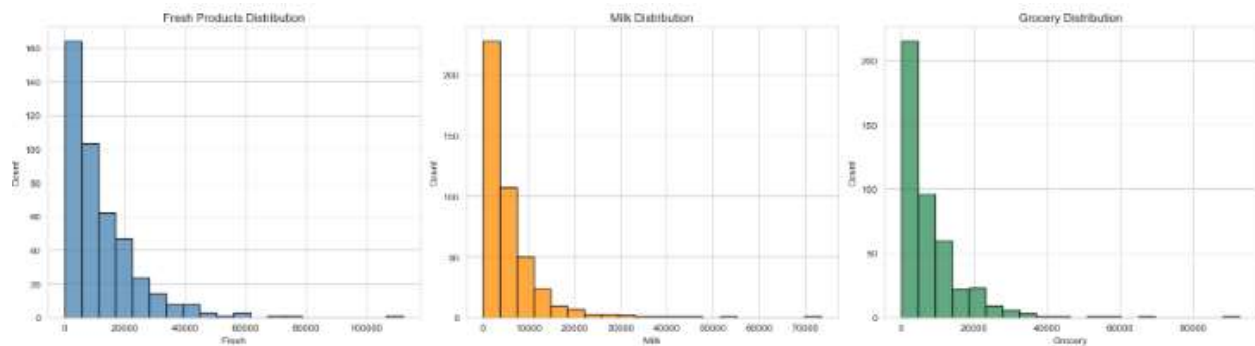
The clustering dataset used in this study is the Wholesale Customers dataset, downloaded via kagglehub from the binovi/wholesale-customers-data-set repository. It contains 440 customer records and 8 columns: Channel, Region, Fresh, Milk, Grocery, Frozen, Detergents_Paper, and Delicassen, where the last six columns record annual spending in monetary units on each product category. The dataset contained no missing values in any column, as confirmed by an explicit null check. The summary statistics below, reproduced directly from the notebook's descriptive statistics output, show that spending is heavily right-skewed: for example, Fresh spending ranges from a minimum of 3 to a maximum of 112,151, with a mean of 12,000.30 and a standard deviation of 12,647.33, indicating a small number of very high-spending customers alongside many lower-spending ones.

Feature	Mean	Std	Min	Max
Fresh	12000.30	12647.33	3	112151
Milk	5796.27	7380.38	55	73498

Feature	Mean	Std	Min	Max
Grocery	7951.28	9503.16	3	92780
Frozen	3071.93	4854.67	25	60869
Detergents_Paper	2881.49	4767.85	3	40827
Delicassen	1524.87	2820.11	3	47943

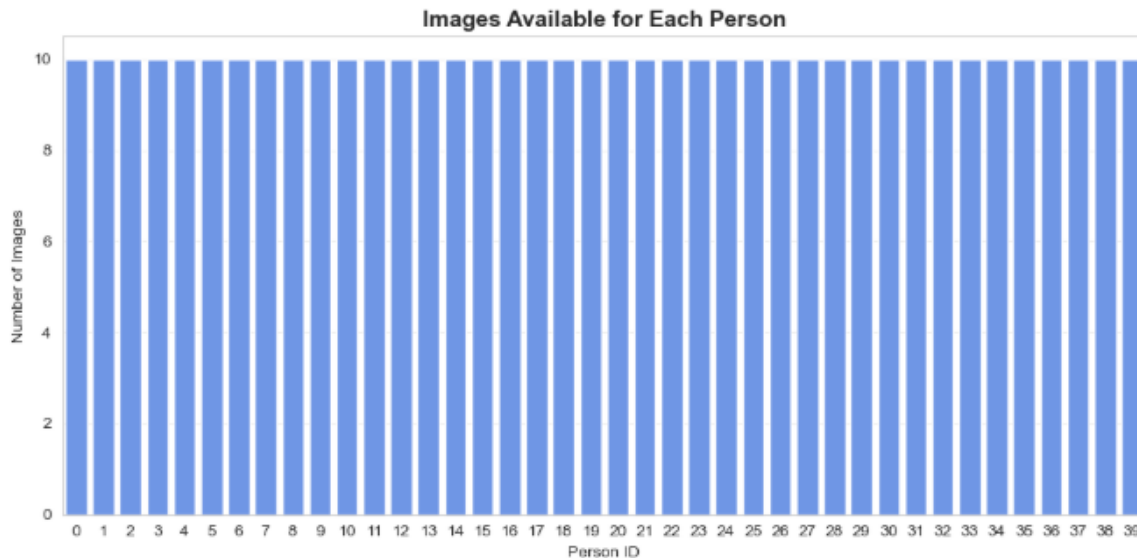
Table 1: Summary statistics for the Wholesale Customers dataset (reproduced from `df1.describe()`).

For the clustering exercise, three of the six spending features were selected: Fresh, Milk, and Grocery. Their distributions, shown below, are all strongly right-skewed with long tails of high-spending outliers, which is consistent with the wide gap between the mean and maximum values reported above.



Olivetti Faces Dataset (High-Dimensional Dataset)

The high-dimensional dataset used in this study is the Olivetti Faces dataset, loaded directly through scikit-learn's `fetch_olivetti_faces` function (a copy was also downloaded via kagglehub but the scikit-learn version was the one actually used for analysis). It consists of 400 grayscale face images belonging to 40 distinct people, with exactly 10 images per person, confirmed by the value counts of the `Person_ID` target. Each image is stored as a flattened vector of 4096 pixel intensities, corresponding to an original resolution of 64 by 64 pixels, with all intensity values normalised between 0 and 1. There were no missing values anywhere in the dataset. With only 400 samples describing each other across 4096 features, this dataset has, on average, roughly 10.24 features per sample, making it a clear example of a high-dimensional, small-sample-size problem where the curse of dimensionality discussed in Section 2.5 is directly relevant.



Part 1: Clustering the Wholesale Customers Dataset

This section applies five clustering algorithms, K-Means, Hierarchical (Agglomerative) Clustering, DBSCAN, Gaussian Mixture Models, and Mean Shift, to the scaled Fresh, Milk, and Grocery features of the Wholesale Customers dataset, tunes their key hyperparameters, and compares their performance using the Silhouette Score, Davies-Bouldin Index, and Calinski-Harabasz Index.

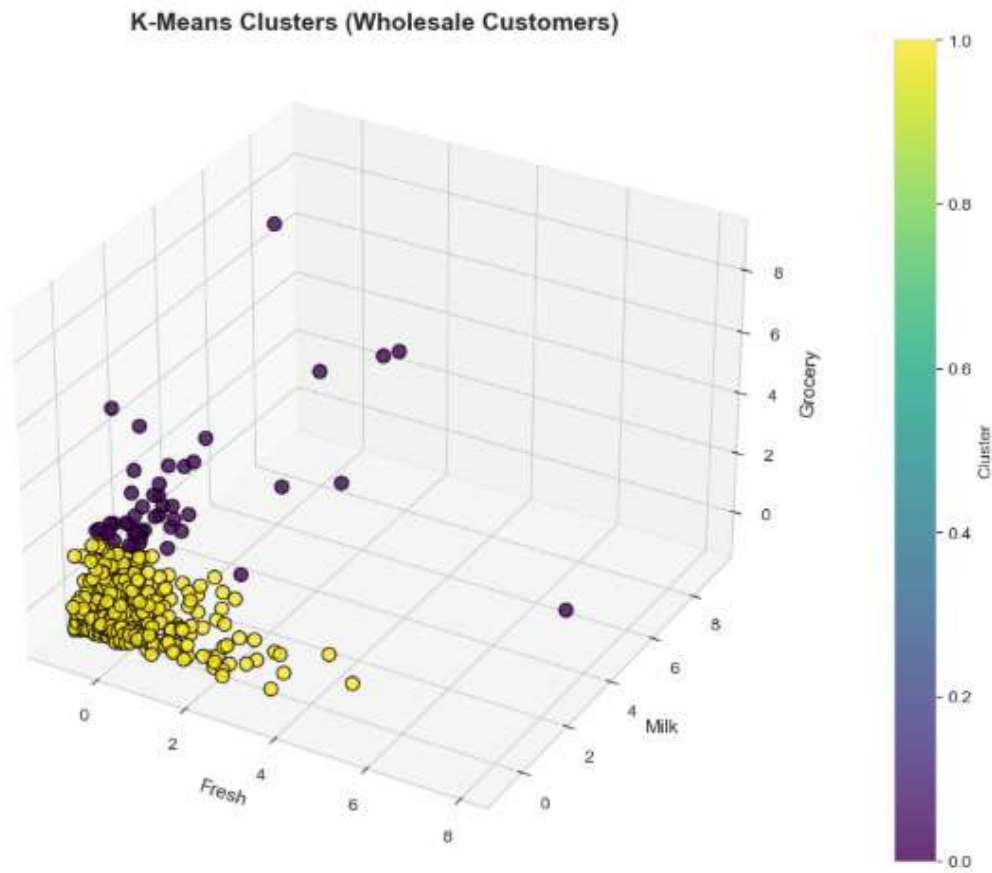
K-Means Clustering

The optimal number of clusters for K-Means was selected by running the algorithm for every value of k from 2 to 10, recording the inertia (elbow method) and the Silhouette Score at each value, and choosing the k with the highest Silhouette Score.



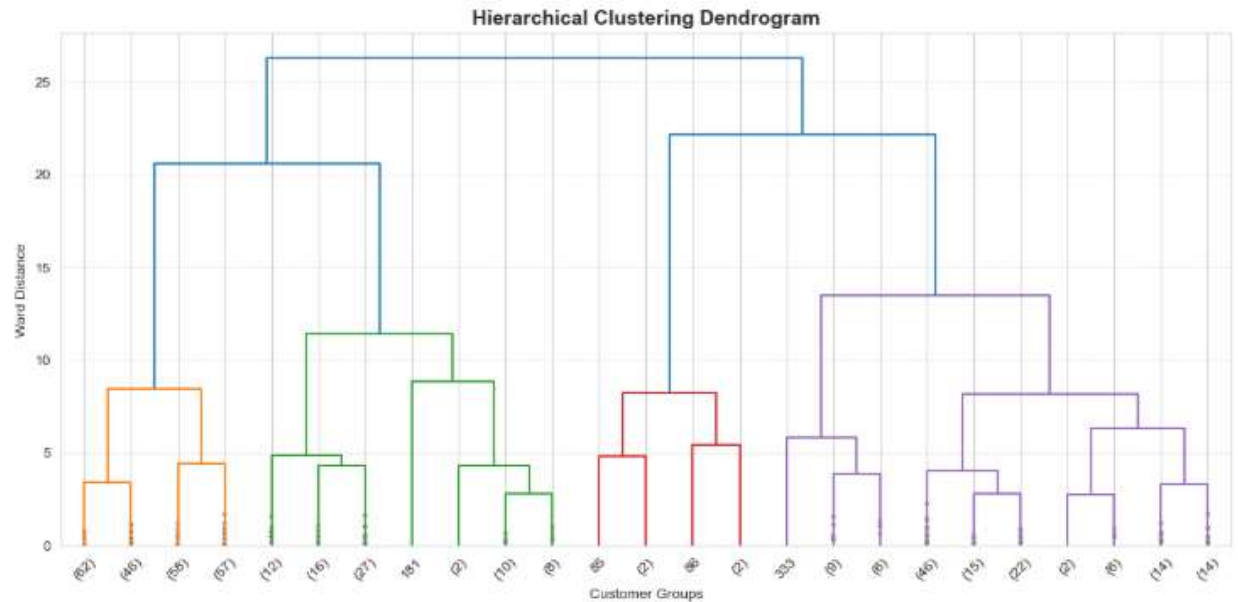
The Silhouette analysis identified $k = 2$ as the optimal number of clusters, and the final K-Means model fitted with this value produced an inertia of 872.38. The resulting clusters, visualised in

three dimensions across the Fresh, Milk, and Grocery axes, form two visually distinguishable groups.



Hierarchical Clustering

Hierarchical clustering was explored using Ward linkage, which merges clusters so as to minimise the increase in total within-cluster variance at each step. A dendrogram, truncated to the last 25 merges for readability, illustrates how individual customers are progressively merged into larger groups.



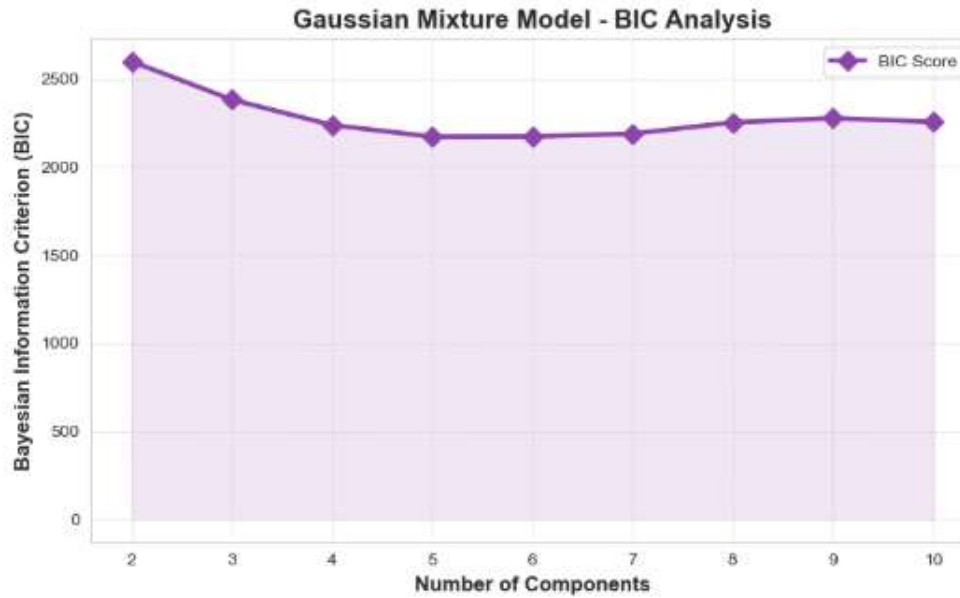
Using the same number of clusters identified as optimal for K-Means ($k = 2$), Agglomerative Clustering was then fitted directly, confirming 2 clusters were produced.

DBSCAN

DBSCAN requires two hyperparameters, `eps` (the neighbourhood radius) and `min_samples` (the minimum number of points required to form a dense region), neither of which can be chosen from a simple elbow-style plot in the same way as K-Means. The notebook therefore performed a grid search over `eps` values from 0.3 to 1.5 in steps of 0.1 and `min_samples` values from 3 to 10, keeping the combination that produced the highest Silhouette Score among all parameter combinations that yielded more than one cluster. The best parameters found were `eps = 0.80` and `min_samples = 6`, giving a Silhouette Score of 0.4551. Fitting DBSCAN with these parameters produced 2 clusters and flagged 21 points as noise (points not assigned to any dense region).

Gaussian Mixture Models (GMM)

For GMM, the number of mixture components was chosen using the Bayesian Information Criterion (BIC), which penalises model complexity while rewarding goodness of fit; lower BIC values are preferred.



The lowest BIC score, 2171.93, was achieved with 5 components, and the final GMM was therefore fitted with 5 clusters rather than 2. This is an important result to note explicitly: BIC is a statistical model-selection criterion that balances fit and complexity for the mixture model itself, and does not necessarily coincide with the cluster count that produces the best-separated groups, as the evaluation in Section 5.6 shows.

Mean Shift

Mean Shift does not require the number of clusters to be specified in advance; instead, it requires a bandwidth parameter controlling the radius of the region considered when estimating the local density mode around each point. The notebook estimated this bandwidth automatically using scikit-learn's `estimate_bandwidth` function with a quantile of 0.25, which produced a bandwidth of 1.1722. Fitting Mean Shift with this bandwidth produced 10 clusters, considerably more than the other algorithms, since Mean Shift with a relatively small automatically-estimated bandwidth tends to find many local density peaks in the data rather than a small number of broad groups.

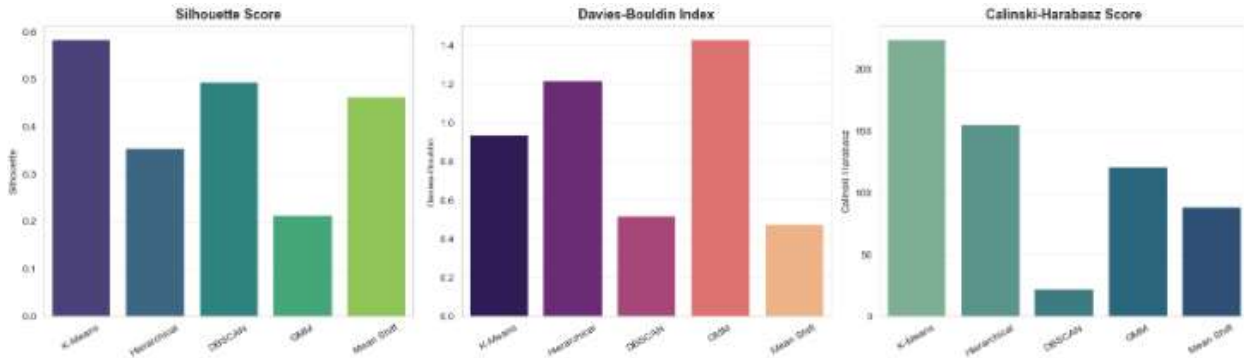
Cluster Evaluation and Comparison

All five clustering algorithms were then evaluated on the same scaled feature set using the Silhouette Score, Davies-Bouldin Index, and Calinski-Harabasz Index. For DBSCAN, the 21 points labelled as noise were excluded from the evaluation, since noise points do not belong to any cluster.

Algorithm	Clusters	Silhouette	Davies-Bouldin	Calinski-Harabasz
K-Means	2	0.5847	0.9412	224.7408
Hierarchical	2	0.3549	1.2197	155.6022

Algorithm	Clusters	Silhouette	Davies-Bouldin	Calinski-Harabasz
DBSCAN	2	0.4951	0.5199	22.7199
GMM	5	0.2139	1.4322	121.3963
Mean Shift	10	0.4633	0.4773	88.7119

Table 2: Comparison of clustering algorithms on the Wholesale Customers dataset.

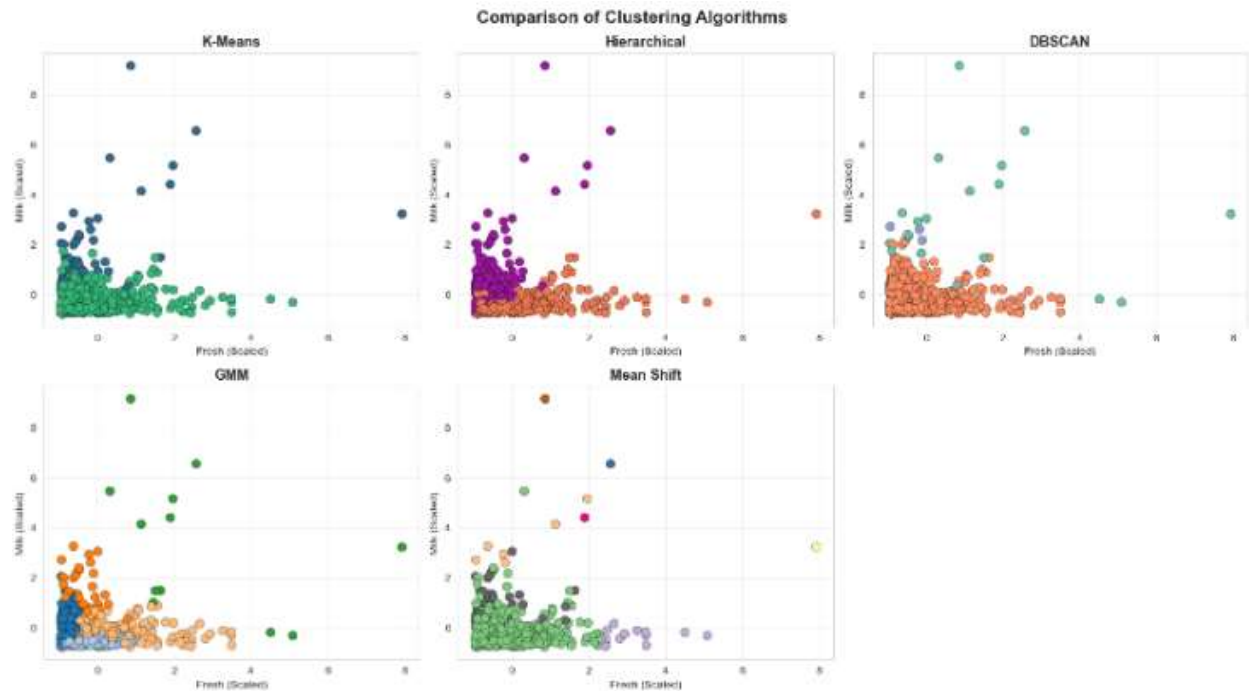


K-Means was identified as the best-performing algorithm, achieving the highest Silhouette Score of 0.5847 and the highest Calinski-Harabasz Score of 224.7408 among all five methods; DBSCAN achieved the lowest (best) Davies-Bouldin Index of 0.5199, closely followed by Mean Shift at 0.4773, but neither matched K-Means on the other two metrics. K-Means performs best here primarily because the underlying spending data, once scaled, forms two reasonably compact and roughly convex groups of customers, an assumption that matches K-Means' own modelling assumption of spherical, similarly-sized clusters particularly well in only three dimensions. Hierarchical clustering, despite using the same target of 2 clusters, scored considerably lower on the Silhouette Score (0.3549), most likely because Ward linkage is sensitive to the presence of outlying high-spending customers, which can distort early merges in the dendrogram and produce less balanced final groups than K-Means' iterative centroid-based refinement. DBSCAN, although finding the same 2 clusters as K-Means, scored lower on Silhouette and considerably lower on Calinski-Harabasz (22.72), because its noise-based mechanism removes ambiguous points rather than optimising an overall compactness objective, and the varying local density of retail spending data means DBSCAN's fixed eps neighbourhood does not capture the two groups as tightly as K-Means' centroid-based partitioning. GMM, despite being a more flexible probabilistic model, produced the worst Silhouette Score (0.2139) once forced into 5 components by BIC; this demonstrates that GMM's likelihood-based, complexity-penalised selection criterion is optimising a different objective (statistical fit of a mixture density) than the geometric compactness and separation that Silhouette and Calinski-Harabasz measure, so the two do not necessarily agree. Mean Shift's automatically estimated bandwidth led it to identify 10 small clusters, which fragments the two natural customer segments into smaller sub-groups and explains its middling

Calinski-Harabasz Score (88.71) relative to K-Means, even though its Davies-Bouldin Index was competitive.

Visualisation of Clustering Results

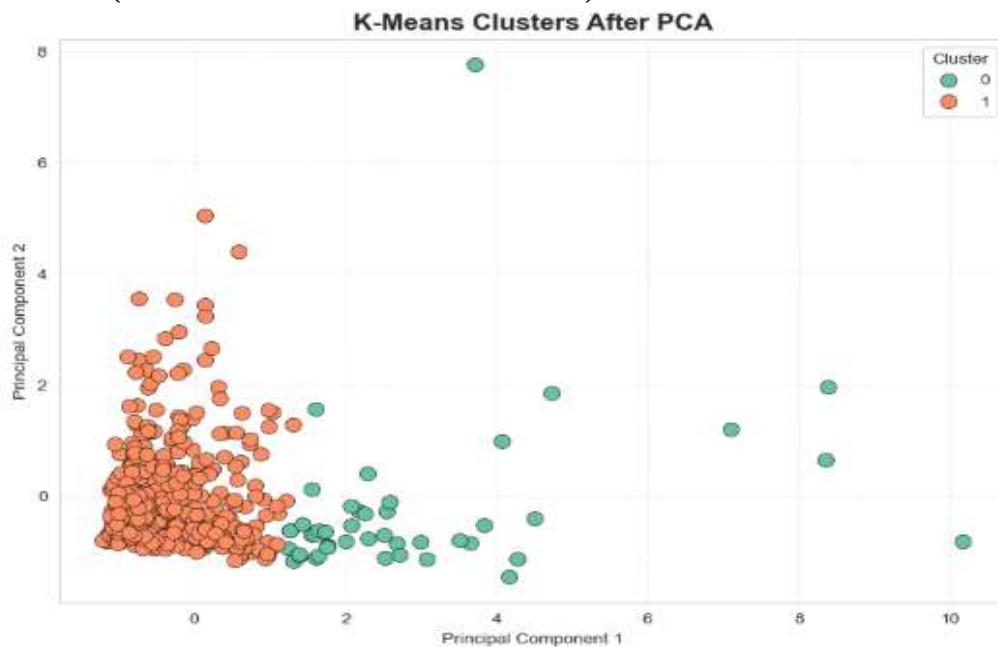
The cluster assignments produced by all five algorithms are shown together below, projected onto the Fresh and Milk axes (both scaled) for direct visual comparison.



Clustering After PCA

To examine whether reducing the three scaled features to two principal components changed the clustering outcome, PCA was applied to produce a 2-dimensional representation of the data, and

K-Means (with the same $k = 2$ identified earlier) was re-fitted on this PCA-reduced representation.



The Silhouette Score achieved on the original 3-dimensional scaled data was 0.5847, compared with 0.5966 after PCA reduction to two dimensions, a small but consistent improvement. Because the original feature space was already low-dimensional (only three features), PCA here mainly removes a small amount of correlated, redundant variance between Fresh, Milk, and Grocery spending rather than solving any serious curse-of-dimensionality problem; the resulting slight improvement in Silhouette Score suggests that the discarded third dimension contributed marginal noise rather than useful separating information for these two groups.

Preprocessing Experiment

The notebook then ran a dedicated experiment to test how the choice of feature-scaling method affects K-Means clustering quality. The first comparison, between `StandardScaler` and `MinMaxScaler`, produced the following Silhouette Scores.

Scaler	Silhouette Score
StandardScaler	0.5847
MinMaxScaler	0.5700

Table 3: Silhouette Score comparison between `StandardScaler` and `MinMaxScaler`.

This comparison was then extended to include `RobustScaler` and the raw, unscaled data, giving a fuller picture.

Scaler	Silhouette Score
StandardScaler	0.5847
MinMaxScaler	0.5700
RobustScaler	0.6411
No Scaling	0.5297

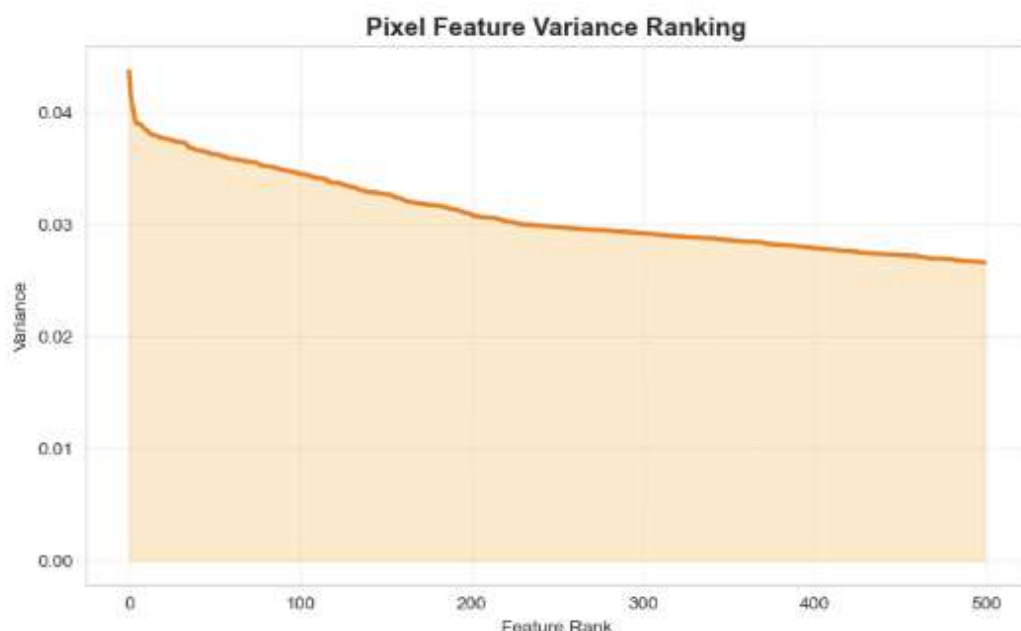
Table 4: Extended preprocessing comparison including RobustScaler and no scaling.

RobustScaler produced the best result of all, with a Silhouette Score of 0.6411, notably higher than StandardScaler. This is consistent with the earlier observation that the spending features are heavily right-skewed with extreme high-spending outliers (recall the large gap between the mean and maximum values reported in Table 1). RobustScaler rescales each feature using the median and interquartile range rather than the mean and standard deviation, making it far less sensitive to these extreme outliers, and so it produces a feature space in which the two natural customer groups are separated more cleanly. At the other end, using no scaling at all produced the lowest Silhouette Score of 0.5297, since without scaling, Grocery and Fresh, which have the largest raw numeric ranges, dominate the distance calculations used by K-Means and distort the clustering relative to Milk.

Part 2: Dimensionality Reduction and Clustering on the Olivetti Faces Dataset

Curse of Dimensionality: Feature Selection

With 4096 raw pixel features describing only 400 images, the Olivetti Faces dataset is a direct real-world instance of the curse-of-dimensionality problem illustrated synthetically in Section 2.5. As a first mitigation step, the notebook ranked all 4096 pixels by their variance across the dataset and retained only the top 2000 highest-variance pixels, reducing the feature count from 4096 to 2000 while keeping the pixels that vary most, and are therefore likely to be most informative, across different faces.



The variance-ranking plot shows a sharp initial decline followed by a long, flat tail, indicating that a relatively small number of pixels, likely corresponding to central facial features such as the eyes, eyebrows, nose, and mouth, account for most of the variability between different people's faces, while the remaining pixels, likely corresponding to background and edge regions, contribute comparatively little discriminative information.

Preprocessing Experiment: Pixel Feature Count

To verify that 2000 was a sensible number of pixels to retain, the notebook compared four different candidate feature counts (500, 1000, 2000, and 3000 top-variance pixels), each scaled, reduced to 2 dimensions with PCA, clustered with K-Means (using the true number of people, 40, as k), and scored with the Silhouette Score.

Top Pixel Features	Silhouette Score
500	0.386059
1000	0.379707
2000	0.382281
3000	0.366373

Table 5: Silhouette Score for different numbers of retained top-variance pixel features.

The Silhouette Score is highest with only 500 pixels and decreases slightly as more, lower-variance pixels are added, falling to its lowest value at 3000 pixels. This pattern reinforces the curse-of-dimensionality intuition from Section 2.5: beyond a certain point, additional pixels contribute mostly noise rather than genuinely useful separating information, so retaining too many low-variance features can slightly dilute cluster quality even after dimensionality reduction. The

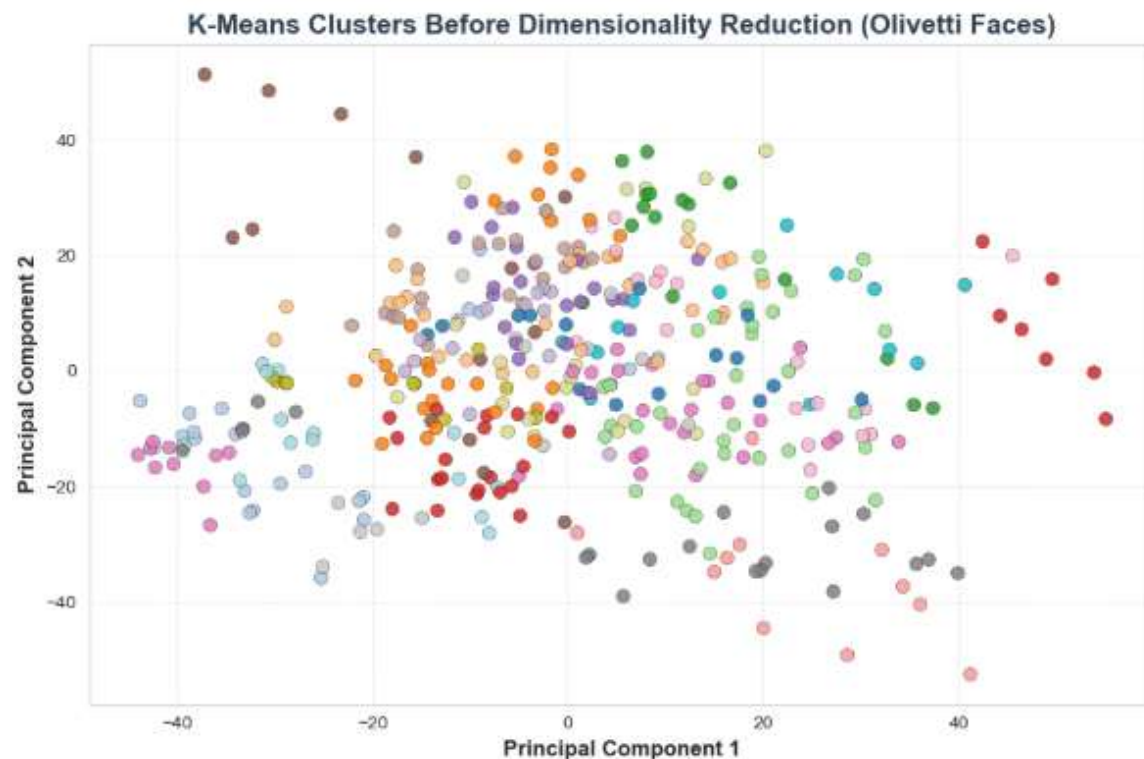
notebook nonetheless proceeded with the top 2000 pixels for the main analysis, which retains a comparable Silhouette Score to the 500-pixel option while preserving more visual information for later interpretation steps such as the feature-importance heatmap in Section 7.

Scaling the Selected Pixel Features

The selected 2000 pixel features were standardised using `StandardScaler`, producing a scaled feature matrix of shape 400 rows by 2000 columns, which was used as the basis for all subsequent dimensionality-reduction and clustering steps on this dataset.

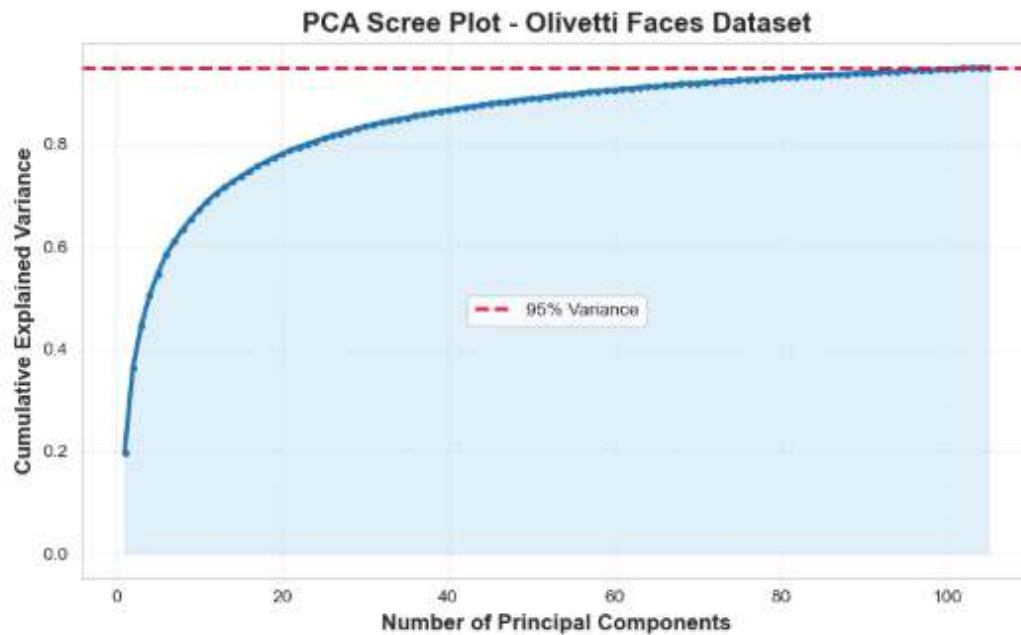
Clustering Before Dimensionality Reduction

As a baseline, K-Means was applied directly to the full 2000-dimensional scaled pixel feature set, using $k = 40$ to match the known number of people. This produced a Silhouette Score of only 0.1352, considerably lower than any of the clustering results obtained on the low-dimensional Wholesale Customers dataset in Section 5. This low score is a direct, empirical demonstration of the curse of dimensionality discussed in Section 2.5: in a 2000-dimensional space with only 400 samples, distances between points lose much of their discriminative power, making it difficult for a distance-based algorithm such as K-Means to find well-separated clusters. For visual reference, the resulting clusters are shown below projected onto the first two principal components (this projection is used purely for plotting; the clustering itself was performed in the full 2000-dimensional space).



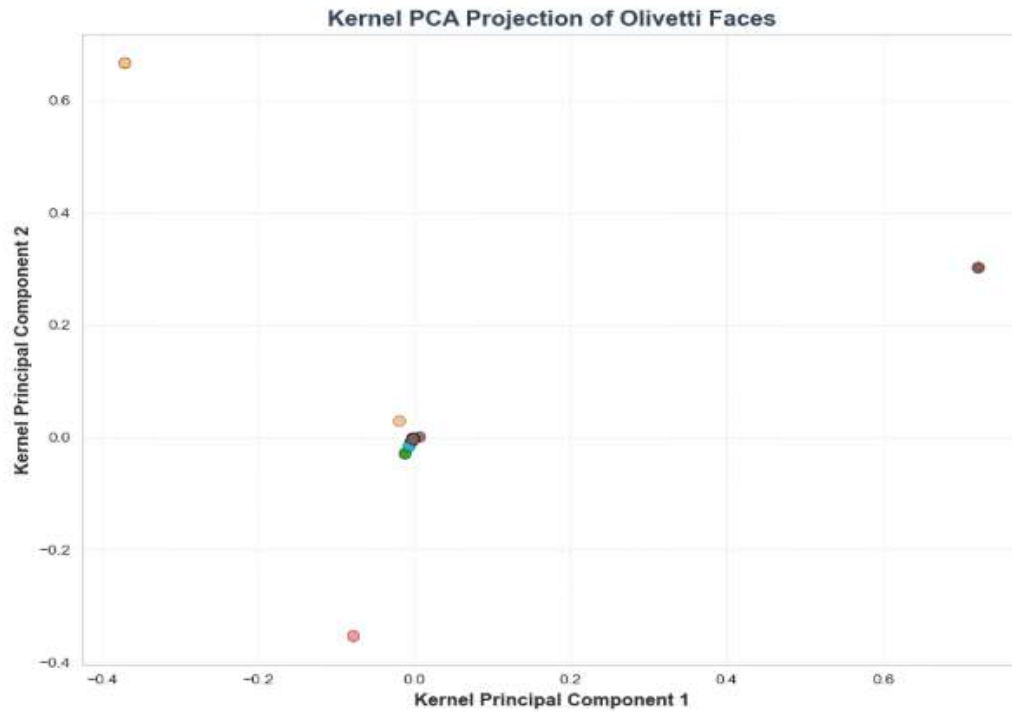
Principal Component Analysis (PCA)

PCA was then fitted to the scaled 2000-pixel data, requesting enough components to explain 95 percent of the total variance. The result was that only 105 components out of the original 2000 features were needed to explain 95.02 percent of the variance, confirming that the pixel data, even after the initial variance-based feature selection, contains substantial redundancy.



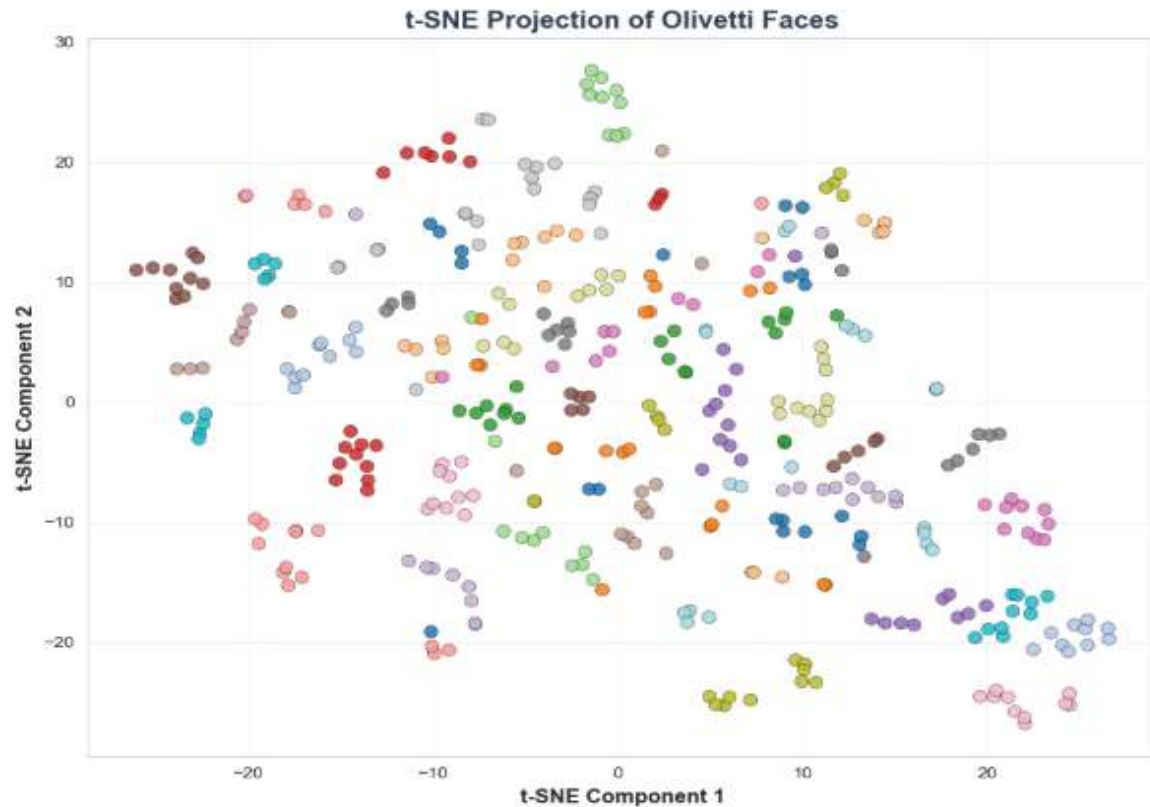
Kernel PCA (RBF Kernel)

Kernel PCA extends PCA by first implicitly mapping the data into a higher-dimensional space using a kernel function, in this case a Radial Basis Function (RBF) kernel with $\gamma = 0.01$, before extracting principal components, allowing it to capture non-linear structure that ordinary PCA cannot. Reducing to 2 components and colouring points by their true Person_ID gives the following projection.

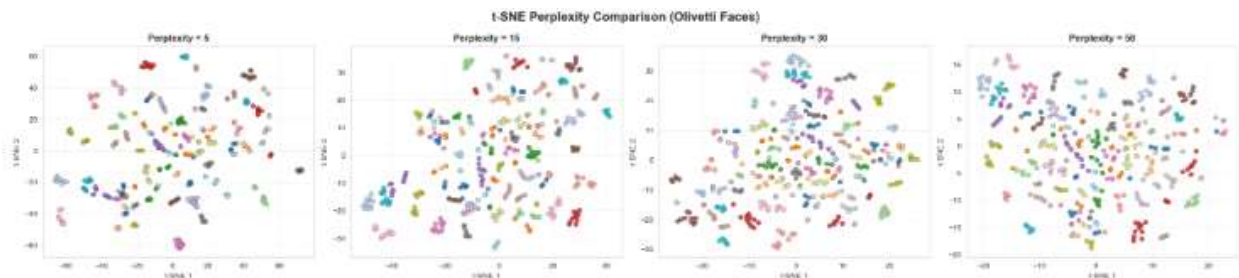


t-SNE

t-SNE was applied with a perplexity of 35, automatic learning rate, and PCA-based initialisation, producing the following two-dimensional embedding, again coloured by the true Person_ID for visual inspection.



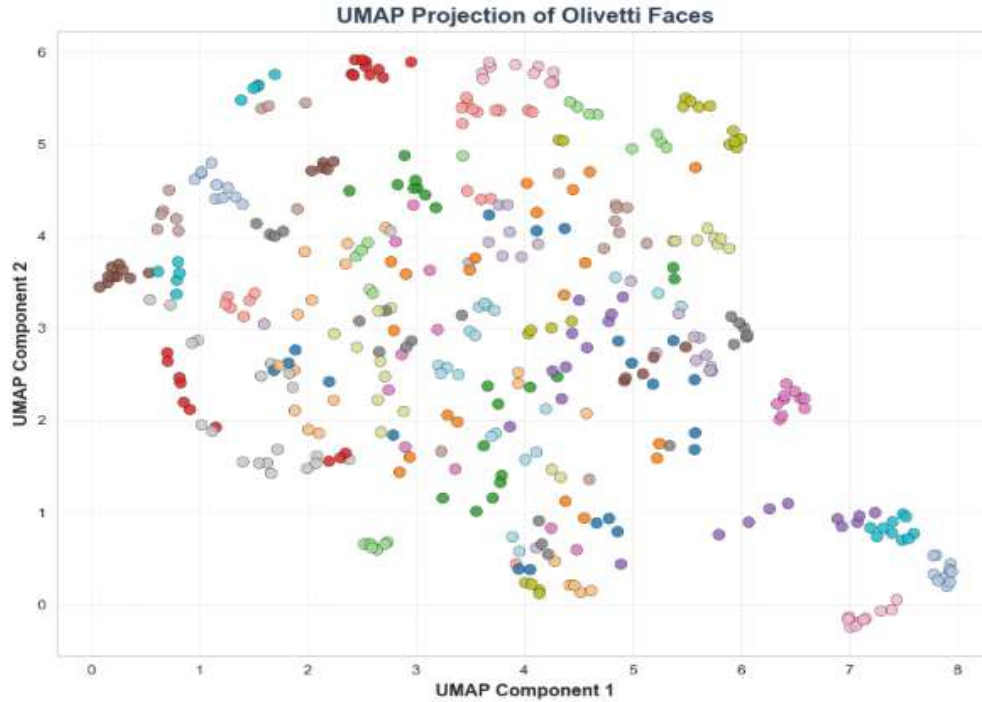
To examine the effect of the perplexity hyperparameter directly, the notebook re-ran t-SNE at four different perplexity values: 5, 15, 30, and 50.



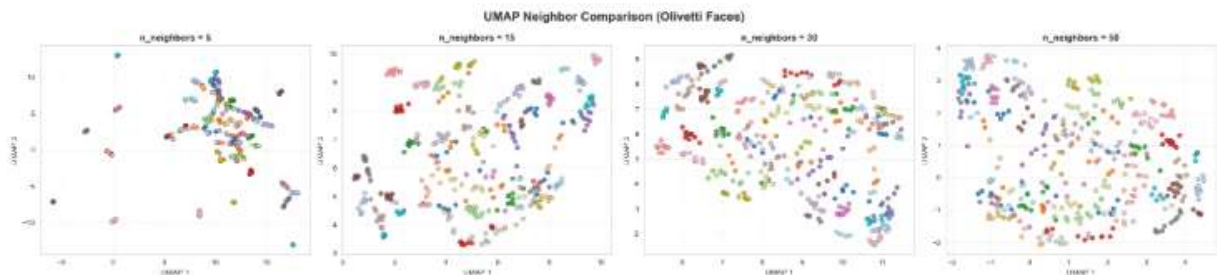
At low perplexity (5), the embedding tends to fragment into many small, tight groups, since only a small local neighbourhood is considered around each point; as perplexity increases toward 30 and 50, more global structure is taken into account and the groups become somewhat larger and less fragmented, though at the cost of some local detail. This illustrates the local-versus-global structure trade-off inherent to t-SNE that was introduced conceptually in Section 2.7.

UMAP

UMAP was applied with 20 neighbours, a minimum distance of 0.15, and the Euclidean metric, producing the following embedding.



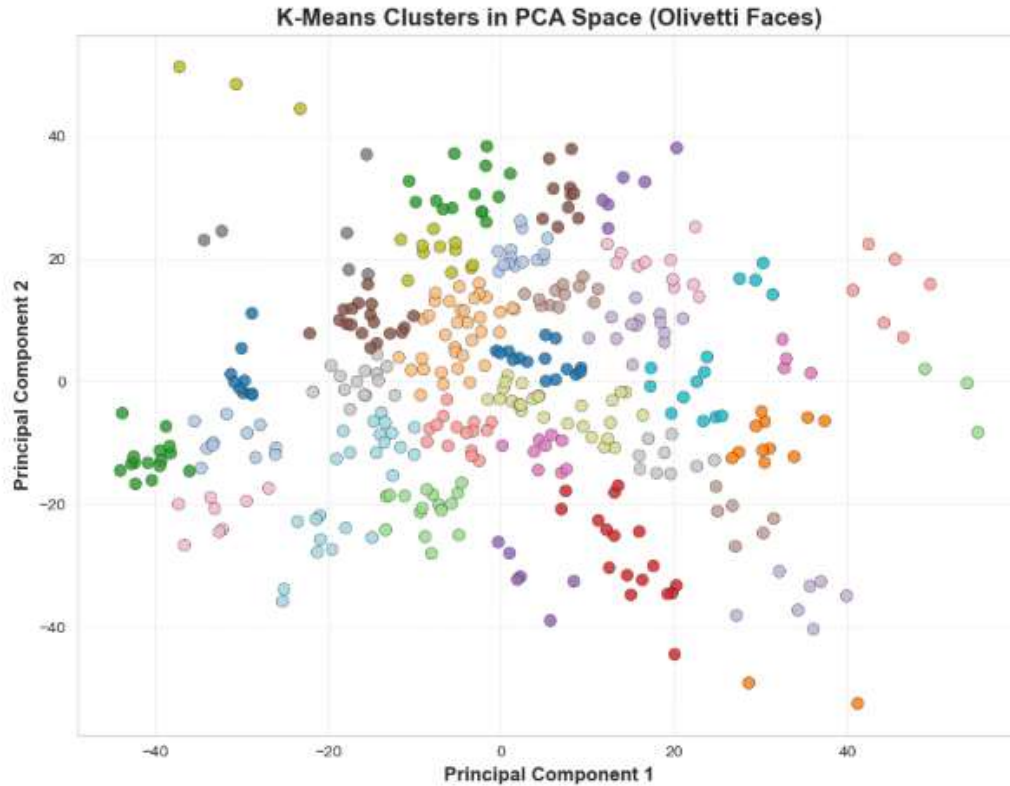
The effect of the `n_neighbors` hyperparameter was then examined directly by re-running UMAP (with the cosine metric) at four different neighbour counts: 5, 15, 30, and 50.



With a small number of neighbours (5), UMAP preserves very fine local structure and produces many small, distinct clusters; as the neighbour count grows toward 30 and 50, the embedding increasingly emphasises global relationships between groups, causing some of the smaller clusters to merge together, mirroring the same local-versus-global trade-off already observed with t-SNE's perplexity parameter.

Clustering on Reduced Data

K-Means, with $k = 40$ to match the number of people, was then applied to a 2-dimensional PCA-based projection of the scaled pixel data, consistent with the projections used for visualisation elsewhere in this section.



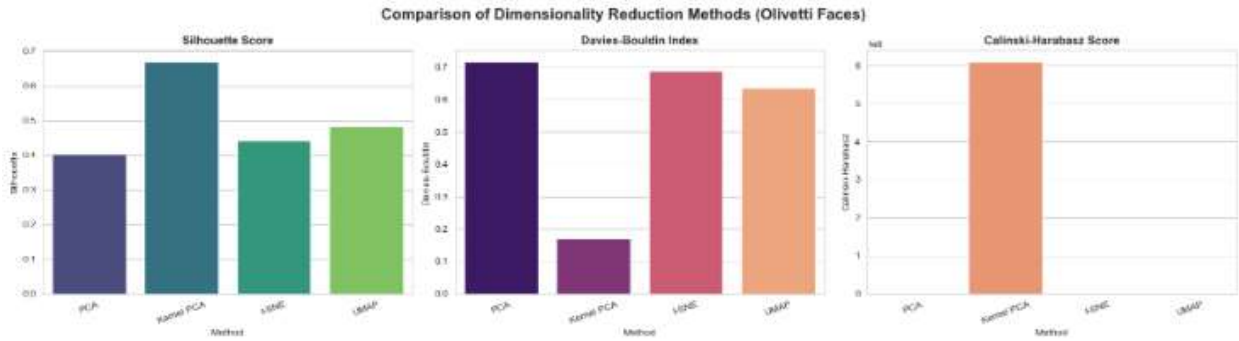
This produced 40 clusters, as expected, with a K-Means inertia of 6761.29.

Compare Dimensionality Reduction Methods

To identify which dimensionality-reduction technique produces the most useful representation for clustering, K-Means ($k = 40$) was applied to the 2-dimensional output of each of PCA, Kernel PCA, t-SNE, and UMAP, and evaluated using the same three metrics used throughout this report.

Method	Silhouette	Davies-Bouldin	Calinski-Harabasz
PCA	0.4031	0.7162	386.36
Kernel PCA	0.6695	0.1707	6.09×10^8
t-SNE	0.4420	0.6881	580.90
UMAP	0.4836	0.6343	762.60

Table 6: Comparison of dimensionality-reduction methods for clustering the Olivetti Faces dataset.



Kernel PCA was identified as the best-performing dimensionality-reduction method, achieving the highest Silhouette Score (0.6695), the lowest (best) Davies-Bouldin Index (0.1707), and an extremely large Calinski-Harabasz Score of approximately 6.09×10^8 . The Silhouette and Davies-Bouldin results both point consistently to Kernel PCA producing the best-separated clusters among the four methods, which is plausible because its RBF kernel can capture non-linear relationships between pixel intensities, for example those arising from variations in pose and illumination across different photographs of the same person, that a purely linear method such as standard PCA cannot represent. The Calinski-Harabasz value for Kernel PCA, however, should be interpreted with some caution: this metric is sensitive to the absolute scale of the projected coordinates, and the particular gamma value used for the RBF kernel can compress or expand the output scale in ways that inflate this specific metric disproportionately; the Silhouette Score and Davies-Bouldin Index, which are scale-invariant or less scale-sensitive, are therefore more trustworthy corroborating evidence that Kernel PCA genuinely produced the best-separated embedding here, rather than the Calinski-Harabasz figure being taken at face value on its own. PCA, by contrast, performed the weakest of the four methods on Silhouette and Davies-Bouldin, consistent with its restriction to linear projections, which cannot fully capture the non-linear variation present in face images. t-SNE and UMAP both performed at an intermediate level; both are designed primarily to preserve neighbourhood structure for visualisation rather than to explicitly maximise cluster separation, which likely explains why they trailed Kernel PCA on these particular metrics despite producing visually plausible groupings in Figures 15 and 17.

Cluster Purity

To check how well the 40 clusters found by K-Means on the PCA-reduced data (Section 6.9) correspond to the true 40 person identities, a cross-tabulation of predicted cluster label against actual Person_ID was produced. An excerpt of this cross-tabulation, covering predicted clusters 0 through 17 against a subset of person identities, shows a mixed pattern: some predicted clusters concentrate most of their images under a small number of true identities, for example predicted cluster 7 places 4 of its images under one particular Person_ID, while other predicted clusters, such as cluster 3, spread their images across several different true identities. This partial correspondence between predicted clusters and true identities is consistent with the moderate Silhouette Scores reported in Table 6 (all in the range of roughly 0.40 to 0.67 rather than close to

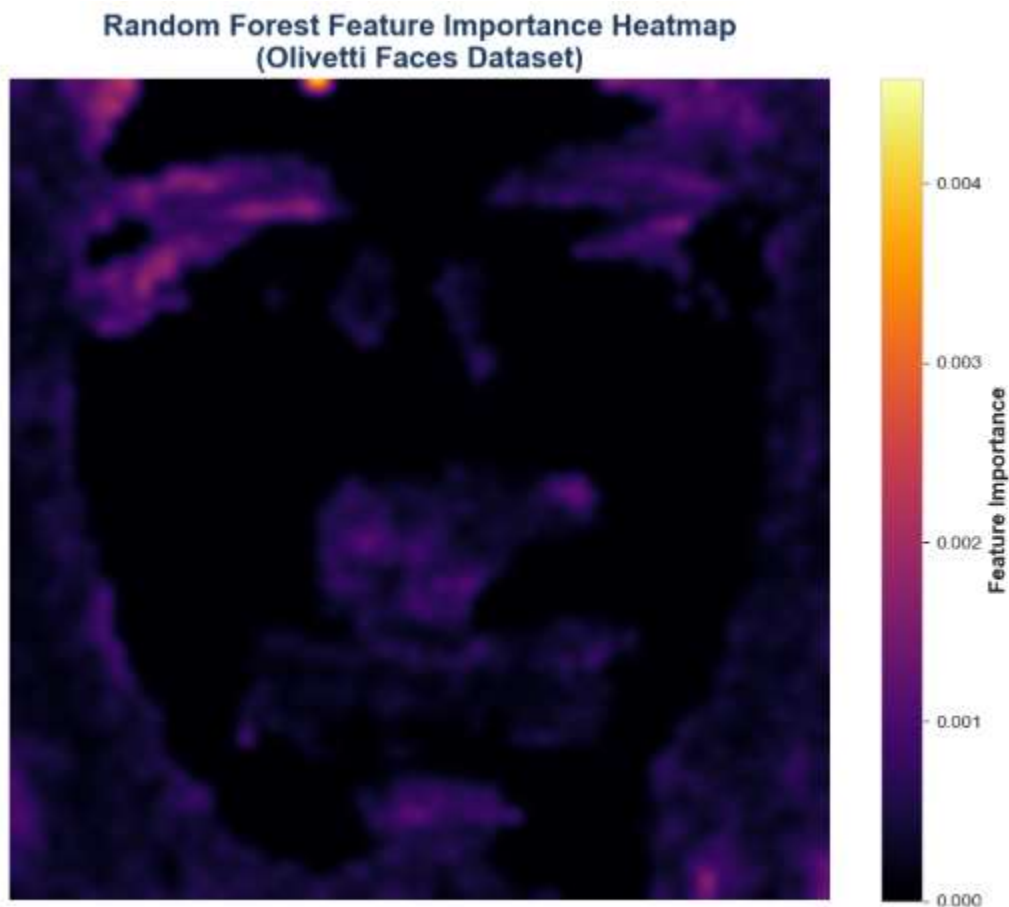
1), and reflects the fact that different photographs of the same person can vary considerably in pose and lighting, causing some individuals' images to be split across more than one predicted cluster, while visually similar faces from different people can occasionally be grouped together.

Model Interpretation

The unsupervised clustering and dimensionality-reduction techniques explored above do not, by themselves, provide an easy way to ask which original pixels most influence a model's ability to distinguish between people. To enable feature-level interpretation, the notebook trained a supervised proxy model, a Random Forest classifier, on the same 2000 selected pixel features, using an 80/20 stratified train-test split. This classifier achieved a test accuracy of 0.95, confirming that the retained pixel features carry strong, learnable signal about a person's identity, and providing a reliable model on which to apply the interpretation techniques described below.

Feature Importance

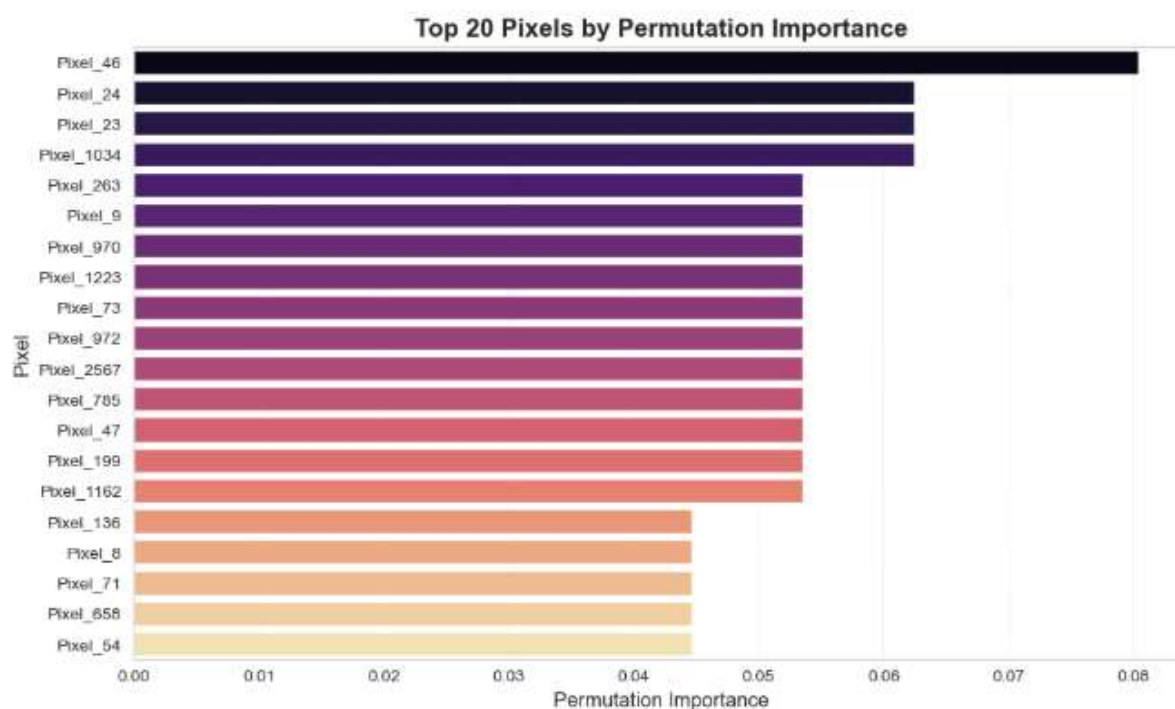
The Random Forest's built-in (Gini-based) feature importances were mapped back from the 2000 selected pixels onto their original positions in the full 64 by 64 pixel grid, with all non-selected pixels set to zero importance, producing the following heatmap.



The heatmap shows importance concentrated in specific regions of the face, most notably around the eyes, eyebrows, and the central nose and mouth area, which are the facial regions that vary most distinctively from one person to another. Background and peripheral pixels show near-zero importance, consistent with the earlier variance-based feature selection in Section 6.1, which had already deprioritised low-variance, low-information pixels in this same region.

Permutation Importance

As a cross-check on the Gini-based feature importances, permutation importance was also computed. Because permutation importance is comparatively expensive to compute across thousands of features, the notebook restricted this analysis to the top 50 most important pixels identified by the Random Forest, retrained a smaller Random Forest on just these 50 pixels, and computed permutation importance with 2 repeats and a 70 percent sample fraction for efficiency.



The pixels identified as most important by permutation importance broadly overlap with those highlighted in the Gini-based feature-importance heatmap in Figure 21, both concentrating around facial regions rather than background pixels. This agreement between two independently computed importance measures strengthens confidence that the highlighted pixels genuinely drive the classifier's ability to distinguish between different people, rather than reflecting an artefact of a single importance calculation method.

SHAP (SHapley Additive exPlanations)

SHAP values were computed using a TreeExplainer applied to the trained Random Forest, for a single test-set face image. The resulting SHAP values were mapped back onto the 64 by 64 pixel

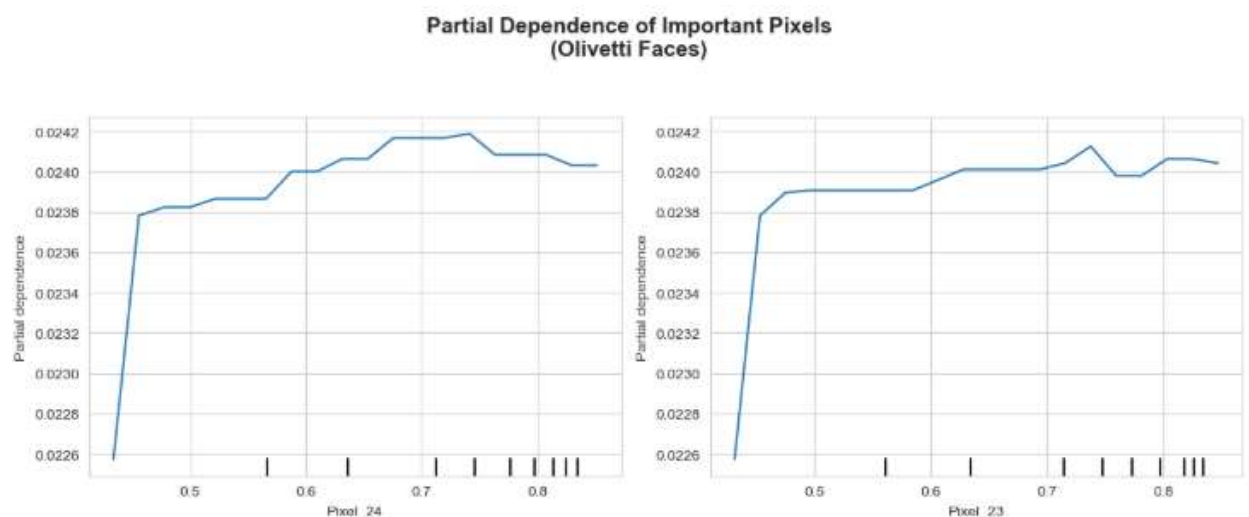
grid and overlaid on the original face image for the class corresponding to the model's actual prediction for that image.



Unlike the global feature-importance heatmap in Figure 21, which summarises importance across the entire training set, this SHAP overlay is a local, sample-specific explanation: the coloured overlay indicates which individual pixels in this particular image pushed the model's prediction toward, or away from, the predicted Person_ID, offering a more granular, case-by-case view of the model's reasoning for that single face.

Partial Dependence Plots (PDP)

Finally, Partial Dependence Plots were produced for the two most important pixel features identified by the Random Forest, showing the predicted probability for a fixed target class (the first class in the model's class list) as the intensity value of each of those two pixels was varied across its range while holding all other pixels at their observed distribution.



These plots make it possible to see whether the model's predicted probability responds smoothly and monotonically to changes in a given pixel's intensity, or whether the relationship is more complex and non-linear; together with the feature-importance and SHAP results above, they complete a picture of both which pixels matter most (feature importance and permutation importance) and how they matter (partial dependence and SHAP).

Conclusion

This study applied a wide range of unsupervised machine learning techniques to two contrasting datasets and used a supervised proxy model to interpret the results. On the low-dimensional Wholesale Customers dataset, K-Means with $k = 2$ was the best-performing clustering algorithm overall (Silhouette Score of 0.5847), and its performance was further improved to a Silhouette Score of 0.6411 by switching from StandardScaler to RobustScaler, which better tolerates the heavy-tailed, outlier-prone spending distributions present in this data. Other algorithms, particularly GMM (whose statistically optimal 5-component solution scored much lower on cluster-separation metrics) and Mean Shift (which fragmented the data into 10 small clusters), illustrate that a criterion used to select a model's internal complexity does not always align with external measures of cluster separation.

On the high-dimensional Olivetti Faces dataset, clustering directly on the 2000 selected pixel features produced a very low Silhouette Score of 0.1352, a direct demonstration of the curse of dimensionality established earlier in the report. Reducing the data to two dimensions before clustering substantially improved cluster quality across all four dimensionality-reduction methods tested, with Kernel PCA (RBF kernel) performing best overall on the Silhouette Score and Davies-Bouldin Index, owing to its ability to capture non-linear structure in the pixel data, such as pose and illumination variation, that linear PCA cannot represent. The moderate cluster-purity results also showed that clustering, even on well-reduced representations, only partially recovers the true 40-person identity structure, reflecting genuine within-person visual variation across photographs.

Finally, the Random Forest proxy model, trained purely to enable interpretation, achieved a strong test accuracy of 0.95 and allowed four complementary interpretation techniques, feature importance, permutation importance, SHAP, and partial dependence plots, to consistently identify the same core facial regions (the eyes, eyebrows, nose, and mouth) as the most informative features for distinguishing between individuals, while also demonstrating how a local, single-sample explanation (SHAP) and a global, dataset-wide explanation (feature importance and permutation importance) can complement one another. Taken together, the results emphasise three recurring themes across both datasets: the choice of preprocessing and scaling method can meaningfully change clustering outcomes; dimensionality reduction is often a practical necessity, not merely an optional visualisation step, when working with high-dimensional data; and model-interpretation techniques provide a valuable bridge back from unsupervised structure to concrete, explainable, feature-level insight.